India's Scanty Fibre and Satellite Infrastructure

In 2019, market analysis group IDATE conducted a survey ranking countries on the basis of homes and buildings connected to fibre (Fibre to the Home [FTTH] and Fibre to the Building [FTTB]). The cut-off criterion was a minimum 1 per cent of homes and buildings connected to fibre. The survey found sixty-four countries in the world meeting this minimum threshold; these countries had 1.2 per cent to 95.7 per cent of homes and buildings connected to fibre. UAE, Qatar, Singapore, China, South Korea, Hong Kong, Japan, Mauritius and New Zealand were the top nine countries with fibre penetration exceeding 53 per cent. Unfortunately, India did not even make the 1 per cent cut, a sad reflection of the state of optical fibre penetration in India.

The IDATE report for Asia Pacific 2020 indicated the same status of India. There were 459 million homes and buildings connected to fibre at the end of December 2019 in the Asia and Pacific. And there were sixteen countries in the Asia and Pacific with more than 1 per cent penetration of FTTH/B. India was not one of these countries. Singapore with 94.8 per cent penetration, China with 80.1 per cent and Hong Kong with 74.9 per cent led the pack. Japan also had a healthy 72.6 per cent penetration rate. In South Asia, Sri Lanka with 1.2 per cent was the only entry in the list.

There are two imperatives that make fibre the choice of digital infrastructure creation. First, the cost of digital services delivered through fibre should be competitive to the cost of digital services delivered through spectrum. Second, it should be possible to make fibre reach homes and buildings.

While the cost of fibre is a pittance, laying fibre in Indian conditions is time-consuming and expensive. Piped natural gas (PNG) had the same disadvantage compared to LPG cylinders. However, two policy decisions smoothened the transition to PNG in Indian cities. First, the transition of buses and other modes to transport to compressed natural gas (CNG) by the